Benchmark Insensitivity to Physical Fidelity: A Simulator-Agnostic Audit of CLEVRER

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Abstract

Counterfactual questions about video, such as "what would happen if this object were removed?", are a standard test of physical and causal reasoning, and CLEVRER is the most widely used benchmark for them. However, a score on this benchmark shows neither which part of a system earned it nor whether a correct answer rests on a real event. In this work, we present CausalVis, a counterfactual engine that answers by intervention. It removes the object, keeps every other object on its recorded path until the change can reach it, hands the reached objects to a simulator, and returns the event that decided each answer. The engine is never trained on questions or answers, and any simulator can be plugged into it. We use it to audit CLEVRER on 1,398 held-out videos. Swapping the simulator across a 2.2× range of rollout error moves the counterfactual score by 2.2 points, and on predictive questions the ranking of simulators is decided by a detector setting rather than by physical accuracy: the most accurate simulator comes last under one detector and first under another. Two choices with no physics in them matter more: when objects are handed to the simulator (1.2 points) and whether the simulation continues past the end of the video (4.6 points). A perfect event detector on the pairs it never simulates is worth 1.4 points, so detection is not the hidden explanation either. Of the 59% of correct answers that can be checked against the annotation, 98.2% agree with it, although half of those checks confirm that a collision did not happen. We also find that CLEVRER's "responsible" labels contradict its own counterfactual answers in 35% of cases. Code, checkpoints and the per-answer records of every reported run are at `https://github.com/bukhari-hamzamukhtar/CausalVis`.

Counterfactual video benchmarks should report the evidence behind each answer alongside its accuracy.

1 Introduction

Reasoning about what would have happened is central to physical understanding [8, 24]. CLEVRER [59] made it a benchmark: 20,000 synthetic videos of colliding objects, with questions such as "What will happen if the cube is removed?". Methods for these counterfactual questions range from neuro-symbolic programs with learned dynamics [14, 59] to differentiable physics [20] and end-to-end transformers [19], and the best system answers 96.1% of options correctly [31].

However, an accuracy number does not tell us which part of a system earned it, or whether a correct answer was reached for the right reason. Recent audits show why this matters. Frontier vision-language models often answer CLEVRER questions correctly without grounding in perception or physics [3], and 35% of the correct answers of mid-tier models rest on physically invalid traces [12]. Audits of other video benchmarks found that single frames or shortcuts explain much of the score [13, 35, 46]. For CLEVRER, Ding et al. [19] observed that about half of the counterfactual questions remove an object that never collides. What the counterfactual score measures has not been tested directly.

In this work, we build CausalVis, an engine that answers counterfactual questions with the three steps of a counterfactual [45]: abduction, action and prediction. The recording plays the role of abduction: every object follows its recorded path until the intervention can reach it. The action is the edit, for example the removal of an object. For prediction, each reached object is handed to a simulator a few frames before it would lose a recorded contact, and the world is simulated past the end of the video. The engine is never trained on questions or answers. The simulator is a module: we use a small learned world model (43k parameters, trained on object motion only), analytic rigid-body laws fitted to training videos, and straight-line motion. Every answer comes with the event that decided it: the two objects, the frame, and whether the event was recorded or simulated. Simulating only the objects an intervention af-

fects was introduced by CRCG [31]; we build on that idea and study when to hand objects over and how far to simulate.

We use the engine to audit CLEVRER's counterfactual questions on 1,398 held-out videos, with every setting chosen on a separate validation split and each test run made once. With the learned world model, CausalVis answers 90.4% of options and 72.3% of questions; with the analytic rigid-body simulator it answers 91.4% and 75.0%. Across four simulators and nine further checkpoints of the learned one, the rollout error changes by a factor of 2.2 and the score by 2.2 points, because four very different simulators give the same answer on 91.4% of options; on predictive questions the ranking of simulators changes with the event detector. The score depends far more on the hand-off timing and on events after the last frame: 34.6% of the questions whose removed object never collides need a collision that happens after the video ends. Of the correct answers whose deciding event can be checked, 98.2% agree with the annotation, and on ComPhy, where the future is annotated, 97.7% of correct predictive answers rest on a real future event. Finally, CLEVRER's explanatory labels often call an object "responsible" for a collision that, according to CLEVRER's own counterfactual answers, still happens without that object (35.3% of cases).

In short, we make the following contributions:

- **An auditable counterfactual engine.** CausalVis answers by explicit intervention and simulation, takes any simulator, needs no question-answer training, and returns the deciding event of every answer (Section 3).
- **A controlled audit of CLEVRER.** With four simulators, nine further checkpoints, one-change ablations, an oracle-detector ceiling and paired tests, we show that neither physical accuracy nor event detection explains much of the counterfactual score, and identify what does (Section 5.2).
- **A measurement of answer validity.** We check the evidence behind each answer against the annotation on CLEVRER, against the annotated future on ComPhy [15] and against the true counterfactual trajectories on CoPhy [6] (Section 5.3).
- **A finding on causal labels.** CLEVRER's "responsible" follows the chain of events, and contradicts CLEVRER's own counterfactual key in 35.3% of the cases where both keys speak (Section 5.4).

All training and evaluation ran on a single laptop CPU. Code, the split and the per-option records will be released.

2 Related work

Counterfactual reasoning about video. CLEVRER [59] pairs synthetic collision videos with descriptive, explanatory, predictive and counterfactual questions. NS-DR [59] and DCL [14] execute question programs on object tracks and roll a learned dynamics model forward. VRDP [20] fits a differentiable rigid-body simulator to each video, ALOE [19] answers with a transformer over object embeddings, and SlotFormer [57] rolls object slots forward for predictive questions. CRCG [31] keeps unaffected objects on their perceived states and simulates only the objects an intervention reaches, which raised VRDP from 94.8% to 96.1% per option. Rovai [47] replays the annotated event log without the removed object and notes that new collisions caused by a removal need a simulator. PhysMind [58] builds one reusable executable world per video from segmentation, meshes and 6D pose, fits analytic dynamics to it, and then inspects, continues or edits it to answer a question, without training. The one-world-per-video design is therefore established and our engine is another instance of it; what we add is an interchangeable simulator and the use of the engine as an instrument, which none of this prior work set out to do. Beyond CLEVRER, ComPhy [15] adds hidden mass and charge, CoPhy [6, 32] asks for counterfactual trajectories under hidden confounders, and CLEVRER-Humans [39] collects human causal judgments. Other benchmarks test causal consistency [2, 10, 22] or physical prediction [9, 11, 53]; Melnik et al. [41] survey them.

Learned physical simulators. Graph-based simulators learn pairwise messages or forces [7, 37, 49]. Energy-based networks conserve energy by construction [18, 25, 48, 54], which makes friction and contact hard to represent [26, 52]; built-in momentum conservation reduces long-horizon error [51] and contact-aware models handle rigid bodies [1]. Inverse graphics recovers physical state by matching a renderer to images [55, 56], and recent work reads rigid-body state and material out of video with a vision-language model [34]. Our learned simulator follows the energy-based line and adds friction by operator splitting [27].

Auditing benchmarks and answers. Buch et al. [13] showed that a model which sees a single frame does well on many video-language tasks; minimal video pairs [35] remove such shortcuts, and Physics-IQ Verified [46] audits a benchmark for video generators [43]. Video generators fail to learn physical laws outside their training data [33], while self-supervised video models show some intuitive physics [23]. Vision-language models answer physics questions without grounding [3, 12, 17]. Grounding language models in a simulator improves physical reason-

ing [5, 16, 38]. We audit both the benchmark and our own answers.

Actual causation. The but-for test calls C a cause of E if E would not have happened without C [36]; structural accounts refine it [28, 29]. People judge causation in collision scenes by simulating the counterfactual world [24], and learned models can support counterfactual inference when the abduction step is tractable [44, 50]. CLEVRER instead labels causes with a rule over the chain of events, and CLEVRER-Humans [39] found that these labels differ from human judgments.

3 CausalVis

Fig. 1 shows the engine on one question, and Fig. 2 shows its parts.

Tracks from video. We use the object detections released with NS-DR [59]: Mask R-CNN [30] masks with predicted colour, material and shape. Detections are linked into tracks by identity. Each mask centre is lifted to the table plane with a camera that we recovered by fitting rendered silhouettes of the three CLEVRER shapes to the masks (median silhouette IoU 0.93). The tracker finds 99.4% to 99.8% of the annotated objects with 99.4% to 99.7% precision on every split, and no annotation is read at test time (Supp. B).

Interventions and reach. Let $x_i(t)$ be the recorded position of object i at frame t, and let $\mathcal{C}$ be the set of objects an intervention changes (for a removal, the removed object). An object is *reached* at the first frame at which one of two things happens. Either it would lose a recorded contact with an object in $\mathcal{C}$ or with an object already being simulated (*lost contact*), or a simulated object touches it (*new contact*). Until then it follows $x_i(t)$ exactly. This is the rule of CRCG [31], applied here to tracks. Following the recording is the abduction step: it carries everything about the scene that the simulator does not model.

Hand-off. An object reached by a lost contact is handed to the simulator L frames before that contact, with the position recorded at that frame and the mean velocity of the last three recorded frames; an object reached by a new contact is handed over at that frame. We use $L = 3$, chosen on validation (Section 5.2). Objects that entered the view fewer than 15 frames earlier are partly outside the frame and appear at about 70% of their speed. For them we divide the velocity by a speed profile fitted on training tracks. The counterfactual world runs E frames past the end of the video ($E = 30$ for the learned model, chosen on validation).

Simulators. We compare four simulators inside the same engine. The *learned world model* predicts a pair energy $V_\theta(x_i - x_j, a_i, a_j)$ from positions and attribute embeddings a_i. The force on object i is

$$F_i = -\sum_{j \neq i} \nabla_{x_i} V_\theta(x_i - x_j, a_i, a_j), \tag{1}$$

computed by automatic differentiation. Because V_θ depends on the difference of positions, $F_{ij} = -F_{ji}$ and momentum is conserved exactly. Motion advances by leapfrog steps wrapped in exact friction half-steps $p_i \leftarrow p_i\, e^{-\gamma_i \Delta t/2}$ (Strang splitting [27]), with a learned friction coefficient γ_i per object. Collisions receive an impulse along the contact normal [42], which is estimated from voxelised object surfaces. The model has 43,460 parameters and was trained on the tracks of 11,182 training videos with a 20-frame rollout loss; it never sees a question or an answer. The *analytic rigid-body* simulator uses constant deceleration for sliding cubes and cylinders, exponential damping for rolling spheres and an impulse along the line of centres, with its six constants fitted on training tracks. *Straight lines* move each object at its hand-off velocity, with or without the fitted friction. *No simulation* deletes the removed object and leaves every other object on its recorded path: its physics is the real physics of the video, and what it lacks is any simulation of the intervention.

Events and evidence. Two objects collide at frame t if their surfaces are closer than a calibrated margin (radii scaled by 1.2, plus 0.03). They also collide if they are within 0.03 and one of them changes velocity by at least 0.002 within 6 frames. An option such as "the cylinder and the sphere collide" is answered yes if the counterfactual world contains such an event, and the answer is flipped for questions asking what will *not* happen. For every option the engine stores the deciding event (the first collision of the asked pair, or its absence) and whether both objects still followed the recording at that frame. This record is the evidence we audit in Section 5.3.

Other question types. For descriptive, explanatory and predictive questions we run CLEVRER's question programs on the recorded events with the semantics of the NS-DR executor, and predictive questions simulate the scene past the end with no intervention. Programs come from templates learned on training questions, which reproduce the dataset's counterfactual programs exactly on the validation split (Supp. B).

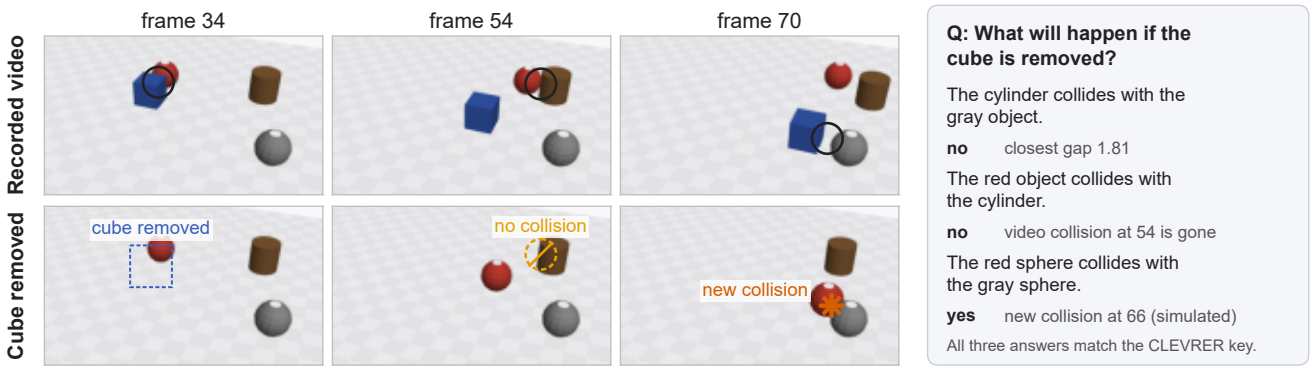


Figure 1: A counterfactual question answered with evidence. The top row shows the recorded video; the bottom row shows the world that CausalVis builds without the cube. Without the cube the red sphere keeps its path, misses the cylinder at frame 54 and strikes the gray sphere at frame 66. Each answer comes with the event that decided it (TEST-A video 10880, question 14).

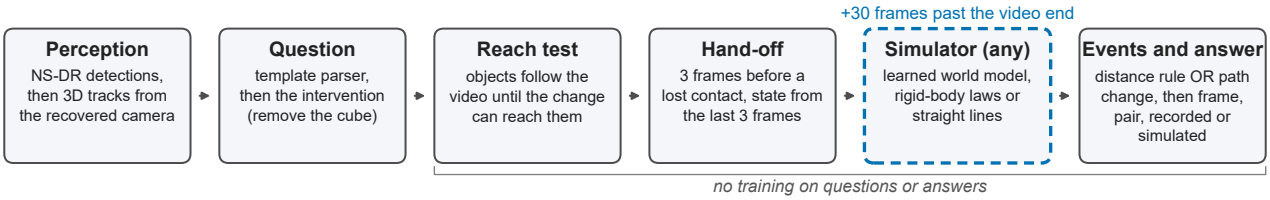


Figure 2: CausalVis. Tracks come from detections and the parsed question sets the intervention. Objects follow the recorded video until the intervention can reach them, and are then handed to a simulator, which can be swapped. The supplement shows the same pipeline running on the example of Fig. 1.

4 Experimental setup

Splits. CLEVRER's test server closed on 16 January 2026 and our requests to the organisers went unanswered, so the answers of its test set are not available to us. Before any experiment we fixed a split of the 15,000 videos that have answers. TEST-A holds 500 CLEVRER validation videos (930 counterfactual questions, 3,332 options) and TEST-B holds 898 CLEVRER training videos that we trained nothing on (1,689 questions, 6,018 options). One caveat belongs here rather than in a footnote: our perception reads NS-DR's detector, which its authors trained on CLEVRER training videos, so TEST-B is held out for our engine and its world model but not for that detector. The two sets score within 0.3 points of each other throughout, which is what we would expect if this did not decide anything, and TEST-A carries no such caveat. VAL-A holds another 500 validation videos (941 questions, 3,391 options) and is used for every choice of setting. The world model and the rigid-body simulator were fitted on 11,182 other training videos.

Protocol. We wrote down the systems and settings to be tested before the test runs (Supp. A). For each simulator, the look-ahead $L \in \{1, 3, 6, 12\}$, the collision detector and the extension $E \in \{0, 10, 20, 30, 45, 60\}$ were chosen on VAL-A by the number of correct options, and each test run was made once. We report accuracy per option and per question (a question is correct when all its options are). Intervals are 95% bootstrap intervals over videos, 2,000 resamples [21]; paired comparisons use McNemar's test [40] on the same options, reported as z ($|z| > 1.96$ is $p < 0.05$).

Physical accuracy. For every simulator we start from true states on 200 VAL-A videos and measure the mean position error after 40 frames, the rollout error. Besides the four simulators we evaluate nine checkpoints of the learned model from earlier runs; six retrained seeds are scored on the benchmark only.

Compute. Every experiment ran on one laptop CPU (Intel i5-8350U, 4 cores, 7.9 GB) without a GPU. Training

Table 1: Counterfactual questions on CLEVRER. Accuracy per option and per question. The three groups are scored on three different sets, so no row is directly comparable across them; the last column gives the fresh TEST-B set alone, which our pre-registration names as primary. †: trained object detector; ‡: attribute labels in training. The rows of CausalVis differ only in the simulator.

Method	Sup.	Option	Question	TEST-B
Published, on CLEVRER's official test set				
NS-DR [59]	†‡	74.1	42.2	
DCL [14]	†	80.4	46.5	
VRDP, no labels [20]	none	89.9	75.7	
ALOE [19]	none	91.4	75.6	
VRDP [20]	†‡	94.8	84.3	
CRCG on VRDP [31]	†‡	96.1	87.8	
PhysMind [58], own 1,000 scenes	none	88.1	70.6	
Ours, on 1,398 held-out videos				
CausalVis, no simulation	†‡	81.4	50.1	81.5 / 50.6
CausalVis, straight lines	†‡	89.7	70.2	89.6 / 70.2
CausalVis, straight + friction	†‡	90.2	71.9	90.2 / 72.2
CausalVis, learned model	†‡	90.4	72.3	90.5 / 72.8
CausalVis, analytic rigid-body	†‡	**91.4**	**75.0**	91.3 / 74.8

the world model took 8.1 hours; answering a counterfactual question takes 2.2 seconds with the learned model and 1.4 with the rigid-body engine, on one thread.

5 Results

5.1 Main results

Table 1 compares CausalVis with published systems. With the learned world model, CausalVis answers 90.4% of options (95% interval 89.7 to 91.1) and 72.3% of questions (70.5 to 74.2). The fitted laws answer 91.4% and 75.0%, 1.0 points above the learned model ($z = 5.0$). The two test sets agree to within 0.4 points for every system (last column of Table 1), so we pool them from here on; every comparison in this section also holds on the fresh TEST-B alone (Supp. J). CausalVis stays 3.4 to 5.7 points per option below VRDP and CRCG, which use the same kind of perception. PhysMind, the closest prior system, reaches 88.1 without any CLEVRER-trained perception, on its own subset, so that difference is not a like-for-like gap in either direction. With no simulation at all the engine already answers 81.4% of options, above NS-DR and DCL; simulating the intervention adds 9.0 points per option and 22.2 per question. For the other direction, a general vision-language model given 16 frames of the video and the same options, Qwen2.5-VL-7B-Instruct [4], answers 54.4% of options on TEST-A, which is what answering "no" to every event scores (54.1%), and 2.9% of questions (Supp. C).

5.2 What the counterfactual score measures

Better physics, same score. Fig. 3a plots the counterfactual score of four simulators and nine further checkpoints of the learned one against their rollout error. The error ranges from 0.050 to 0.113, a factor of 2.2, while the score ranges from 89.2% to 91.4%. The two move in the expected direction ($r = -0.57$, 95% CI $[-0.85, -0.03]$), so the question is not the sign but the size. Nine of the thirteen points are checkpoints of one model, so we read the fitted slope rather than the correlation: 19.4 points per unit of rollout error, which over the observed range is 1.2 points, against 4.6 for a single choice about how long to simulate. Straight lines have 1.8 times the rollout error of the learned model and score only 0.75 points lower. Earlier checkpoints of the learned model range from 89.2% to 90.9%, a spread as large as most changes of simulator. On predictive questions the relation reverses (Fig. 3b): straight lines answer 91.4% of options and the rigid-body engine, which has the lowest rollout error, answers 88.1% ($z = -3.8$ against the learned model). That reversal belongs to the detector, not to the simulators: over fifteen common settings the rigid-body engine comes last in all five that use the plain distance rule and in none of the ten that use the calibrated rule, where it is often first (Supp. D, alike on VAL-A and test). A choice with no simulation in it orders the predictive scores. A physical simulator does not guarantee a better score on either question type. The claim is about accuracy above a low bar and not about physics as a whole: removing the simulation entirely costs 9.0 points per option, while making it 2.2 times more accurate buys 2.2. It is also a claim about a regime. Held at one common hand-off and extension on VAL-A the four simulators differ by 1.5 to 3.5 points depending on the setting, against 2.2 when each is tuned to its own, so tuning does not explain the result (Supp. D). The spread widens as objects are handed over earlier: the benchmark becomes more sensitive to physics the more of the world a system must simulate, and the pipeline that simulates least is the one that scores best. The reason is visible in the answers themselves. The four simulators give the same answer on 91.4% of the 9,350 counterfactual options, although at least one asked object is simulated before the video ends on 47.8% of them. Every difference between simulators comes from the remaining 8.6%, and there the rigid-body engine is right on 67.5% of options, the learned model on 55.7% and straight lines on 47.0%. A yes-or-no question about a collision changes only when a trajectory error moves a pair across the contact distance, and this agreement suggests that most asked pairs are far from that boundary.

What the score depends on. Fig. 4 removes one part of the system at a time. Stopping the simulation at the last

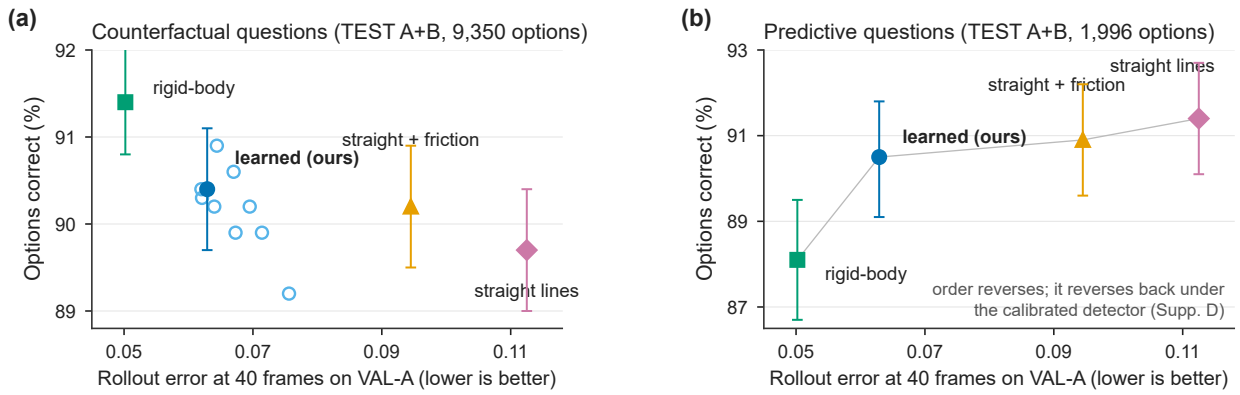


Figure 3: Physics quality against benchmark score. (a) On counterfactual questions a 2.2× spread in rollout error moves the score by 2.2 points, and most differences sit inside the 95% intervals. (b) On predictive questions the order reverses under the detector chosen on validation and reverses back under the calibrated one (Supp. D): the ranking tracks the detector, not the physics. Hollow circles in (a) are nine further checkpoints of the learned model, so the thirteen points are not independent simulators; the fitted slope is 19.4 points of score per unit of error, which over the observed range is 1.2 points. The no-simulation baseline (81.4) is below the axis. Error is measured on 200 VAL-A videos; scores on TEST-A and TEST-B.

frame costs 4.6 points per option and 14.3 per question. Handing objects over 12 frames before the lost contact instead of 3 costs 1.2 and 3.2 points, and on VAL-A every simulator improves as the hand-off moves later (Supp. D). The contact impulse is worth 0.8 points and the path-change detector 0.3. Removing the learned pair energy, so that objects interact only through contact impulses, improves the score by 0.4 points; the learned friction and the entry correction make no measurable difference. The hand-off and the extension decide when simulation starts and how long the world lasts, and neither involves physics.

A perfect event detector. More of our checkable errors come from event detection than from the simulator (Section 5.3), so we priced it. On 52.2% of options neither asked object is ever simulated, so the counterfactual world equals the recording there and a perfect detector is CLEVRER's annotation, which the engine never reads. Substituting it raises the learned system from 90.4% to 91.8% and the rigid-body simulator from 91.4% to 92.9% ($z = 9.8, 10.2$): perfect detection on half the benchmark buys about what the whole 2.2× accuracy sweep buys, and the ceiling is still 3.2 points below CRCG. With the oracle blind to everything after the last frame it scores 88.2%, *below* the engine ($z = -8.1$) – the next result, with no model in it.

Collisions after the video ends. Following ALOE [19], we split questions into those whose removed object never collides (48.2%), those whose answers the recorded video already contains (22.9%), and the rest (28.9%), which need counterfactual reasoning. On the last group no simulation answers 5.3% of questions, the learned model 56.6% and the rigid-body simulator 63.2% (Supp. E). In the first group nothing in the video changes, yet the answer key says that a pair collides on 474 options where the video shows no collision of that pair. These options occur in 34.6% of the questions of the group and can only be answered by simulating past the last frame, which explains why the extension matters. Of the learned system's 896 wrong options, 40% invent a collision inside the video and 34% miss one the key places after it ends; only 10% miss a collision the video shows (Supp. E).

5.3 Are the answers right for the right reason?

Each answer of CausalVis names its deciding event. When both asked objects still follow the recording at that event, the event can be compared with CLEVRER's collision annotation, which the engine never reads. Fig. 5a shows the result for the learned system. Of the 8,454 correct answers, 4,973 (58.8%) can be checked this way; they are by construction the ones no simulator touched. Of these, 4,881 (98.2%) agree with the annotation: an annotated collision of the same pair within 15 frames, or no collision of the pair in the whole video. With a 6-frame tolerance the rate is 93.4%, because our detector fires a median of 4 frames before the annotated contact (Fig. 5b). Another 40.5% of correct answers are decided in the simulated part of the world, for which CLEVRER has no ground truth. Bourigault [12] reports 35% of mid-tier vision-language answers resting on invalid traces; our checkable rate is 1.8%, though the two

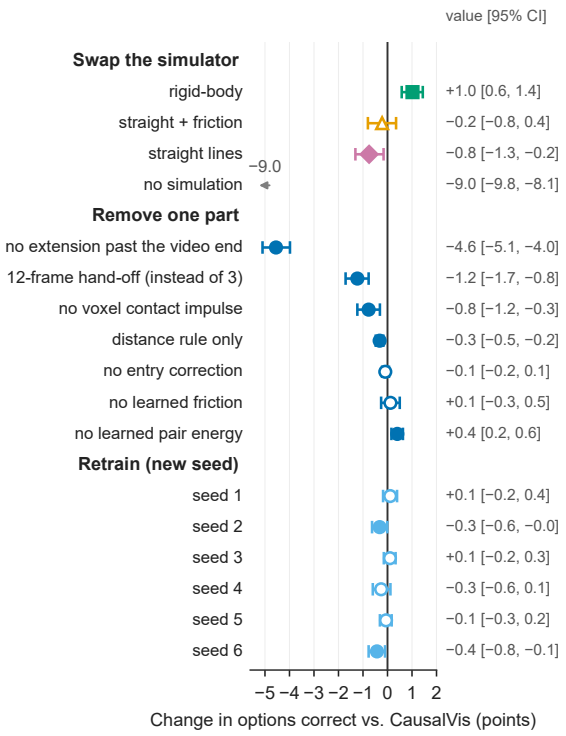


Figure 4: Effect of each change on the full system. Points are paired differences in options correct on TEST-A and TEST-B with 95% video-bootstrap intervals; filled markers exclude zero. One-change intervals are computed within a single checkpoint, so effects below the 0.5-point seed spread need not survive a retrain. The extension past the video end and the hand-off timing matter most; the learned parts matter little, and removing the learned pair energy helps.

checks verify different things.

The evidence also explains errors. Of the 305 wrong answers that can be checked, 188 come from a collision that the detector reports and the annotation does not contain, and 37 from an annotated collision that the detector misses. In 72 more, the key places a collision after the end of the video and our extension does not reach it. On the recorded videos, the event detector used for the other question types finds 95% of the annotated collisions with 79% precision (Supp. F).

CoPhy [6] provides the true counterfactual trajectory, so there the simulated events can be checked too. We run the engine with the rigid-body simulator on BallsCF, infer the hidden masses and restitutions from the observed run, and compare the collisions in the predicted counterfactual run with the true ones (same pair, within one step). With 2 balls, 85% of the reported collisions are real and 65% of the real ones are found; with 4 balls, 61% and 49%, and with 6 balls, 49% and 41% (Fig. 5c). Longer chains of collisions leave more room for a simulated event to drift, which is the same weakness that the counterfactual questions of CLEVRER expose.

ComPhy [15] annotates what happens after the end of each video, so there every predictive answer can be checked. On 400 validation scenes, 1,378 of the 1,411 correct predictive answers (97.7%) rest on a real future event: a collision of the same pair within 5 frames of the simulated one, or no collision of the pair at all. The first simulated collision of an asked pair is real in 93.5% of cases, and 76.9% of the real ones are found.

5.4 What "responsible" means

Our explanatory score (Table 2) is below ALOE's and VRDP's, so the argument below rests on a comparison between CLEVRER's own two answer keys, with no model in it. CLEVRER's explanatory questions ask which event is "responsible" for a collision. The answer key follows a chain rule: an event is responsible if it precedes the collision in the chain of events of the objects involved [59]. The but-for test, the standard test in law and philosophy, reads it differently: an event is a cause when the target would not have happened but for that event [24, 36]. We apply it by removing what the candidate cause adds, simulating that world with the engine, and checking whether the target collision still happens. Both rules can be run on the same options. On TEST-A and TEST-B, the chain rule on our recorded events answers 88.4% of options and 79.7% of questions, and the but-for test 77.7% and 61.5% ($z = -27.2$).

This gap is not only a simulation error. To check this without any model, we paired explanatory and counterfactual options from the same video. The explanatory option states that the presence (or entrance) of Z is responsible for the collision of X and Y, and the counterfactual option asks whether X and Y collide without Z. Over VAL-A, TEST-A and TEST-B, the key calls Z responsible in 320 such pairs. In 113 of them (35.3%, Fig. 6a), CLEVRER's own counterfactual answer says that X and Y still collide without Z. Where the key calls Z not responsible, the two answers disagree in 1 of 1,366 pairs. The counterfactual answer does not give the time of the collision, so a later collision of the same pair could explain some of these cases. It explains few of them: in all 113 cases the pair collides only once in the recorded video, and in the engine's world without Z the pair collides within 10 frames of that recorded collision in 56 cases.

On human judgments the two readings are closer (Fig. 6b). CLEVRER-Humans [39] asks people which events cause which. On it, the but-for test with CausalVis answers 64.4% of options and the chain rule on the same events 60.9%; the best published model answers 54.0% and

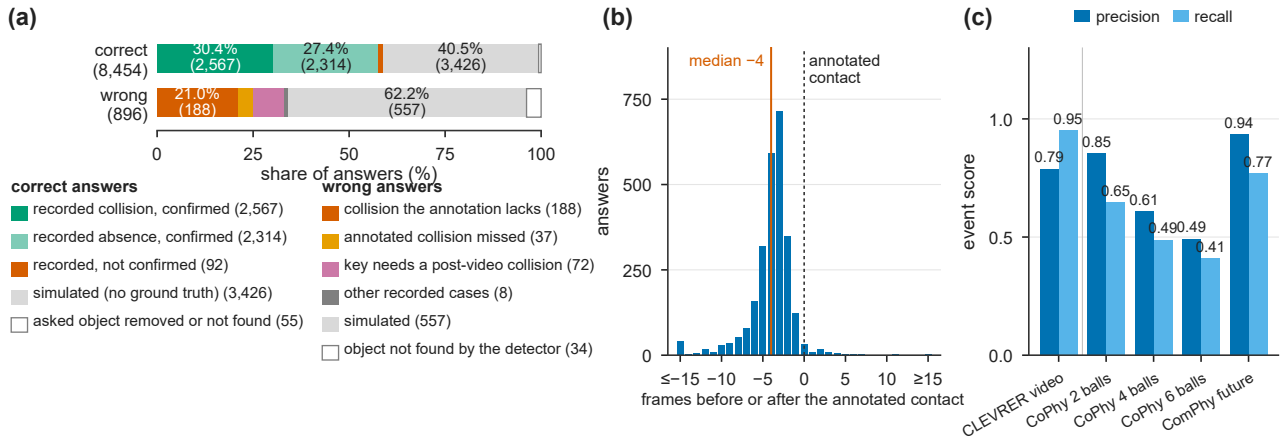


Figure 5: The evidence behind each answer. (a) Most correct answers rest on an event that can be checked against CLEVRER's annotation, and 98.2% of those checks pass; the rest are decided in simulation, where no ground truth exists. (b) The deciding collision is found a median of 4 frames before the annotated contact. (c) Reported events against true events: recorded CLEVRER videos, simulated CoPhy counterfactuals with 2 to 6 balls, and simulated ComPhy futures.

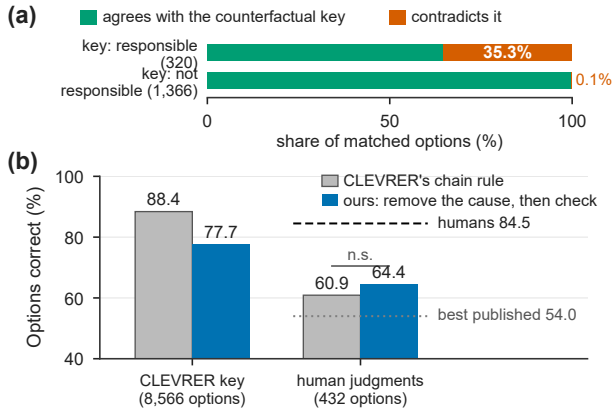


Figure 6: Two meanings of "responsible". (a) When CLEVRER's key marks an object responsible for a collision, its own counterfactual answers say the pair still collides without that object in 35.3% of cases. (b) CLEVRER's chain rule (grey) matches CLEVRER's key. On human judgments the two rules are within noise of each other (54 against 39 discordant options, $z = 1.56$).

people 84.5%. Paired on the same options that margin is not significant (54 against 39 discordant, $z = 1.56$), so the argument rests on the disagreement between CLEVRER's two keys, not on this.

5.5 Other question types and benchmarks

Table 2 reports the other CLEVRER question types. CausalVis is on par with DCL and below ALOE and VRDP.

Table 2: Other CLEVRER question types, per option and per question. Published numbers are on the official test set; ours on TEST-A and TEST-B.

Method	Descr.	Explanatory		Predictive	
		Opt.	Ques.	Opt.	Ques.
NS-DR [59]	88.1	87.6	79.6	82.9	68.7
DCL [14]	90.7	89.6	82.8	90.5	82.0
ALOE [19]	94.0	98.5	96.0	93.5	87.5
VRDP [20]	93.4	96.3	91.9	95.7	91.4
CausalVis	90.9	88.4	79.7	90.5	82.6

Its predictive score uses the learned model, which was best on VAL-A; on the test sets straight lines scored 0.9 points higher.

On ComPhy [15], where masses and charges are hidden, we infer them from the target video and its four reference videos and simulate with fitted laws. With tracks taken from the annotation, the engine answers 79.5% of counterfactual and 86.9% of predictive options on the validation set, against 68.3% and 75.3% for CPL and 80.0% and 88.0% for people on the test set (Supp. H). On CoPhy BallsCF with 2 to 6 balls, its counterfactual position error is 0.89 to 2.60, between 12% and 45% of the error of copying the observed trajectory (Supp. I). Both use oracle tracks and show that the engine transfers; they are not comparable with methods that work from pixels.

6 Limitations

All scenes are synthetic, with three shapes, one size and almost no occlusion, and we did not test the engine on real video. Our perception uses NS-DR's detector, trained with CLEVRER's masks and attributes, so the fair comparison group in Table 1 is VRDP and CRCG. The 40.5% of correct answers decided in simulation can be inspected but not verified, and half of the checks we can make confirm an absence rather than an event. The simulators are planar and model neither toppling nor rotation. CLEVRER's test server had closed, so our test sets are held-out validation and training videos and the comparison with published numbers is approximate. Finally, training on one CPU limited how often we could repeat it: seven checkpoints of one recipe span 90.0 to 90.5 options correct (Supp. C), a spread as large as several of the effects we report, which is itself part of the finding.

7 Conclusion

Used as an instrument, CausalVis shows that CLEVRER's counterfactual score depends little on physical accuracy or event detection, and much on when simulation starts and how long it runs. Such benchmarks should report how many answers physics decides, and how many rest on valid evidence.

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References

[1] Kelsey R. Allen, Yulia Rubanova, Tatiana Lopez-Guevara, William F. Whitney, Alvaro Sanchez-Gonzalez, Peter Battaglia, and Tobias Pfaff. Learning rigid dynamics with face interaction graph networks. In *International Conference on Learning Representations (ICLR)*, 2023.

[2] Tayfun Ates, M. Samil Atesoglu, Cagatay Yigit, Ilker Kesen, Mert Kobas, Erkut Erdem, Aykut Erdem, Tilbe Goksun, and Deniz Yuret. CRAFT: A benchmark for causal reasoning about forces and interactions. In *Findings of the Association for Computational Linguistics: ACL 2022*, 2022.

[3] Ieva Bagdonaviciute and Vibhav Vineet. Physics knowledge in frontier models: A diagnostic study of failure modes. *arXiv preprint arXiv:2510.06251*, 2025.

[4] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibo Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, Xi Zhang, Tianbao Xie, Zesen Cheng, Hang Zhang, Zhibo Yang, Haiyang Xu, and Junyang Lin. Qwen2.5-VL technical report. *arXiv preprint arXiv:2502.13923*, 2025.

[5] Vahid Balazadeh, Mohammadmehdi Ataei, Hyunmin Cheong, Amir Hosein Khasahmadi, and Rahul G. Krishnan. Physics context builders: A modular framework for physical reasoning in vision-language models. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025.

[6] Fabien Baradel, Natalia Neverova, Julien Mille, Greg Mori, and Christian Wolf. CoPhy: Counterfactual learning of physical dynamics. In *International Conference on Learning Representations (ICLR)*, 2020.

[7] Peter Battaglia, Razvan Pascanu, Matthew Lai, Danilo Jimenez Rezende, and Koray Kavukcuoglu. Interaction networks for learning about objects, relations and physics. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2016.

[8] Peter W. Battaglia, Jessica B. Hamrick, and Joshua B. Tenenbaum. Simulation as an engine of physical scene understanding. *Proceedings of the National Academy of Sciences*, 110 (45):18327–18332, 2013.

[9] Daniel M. Bear, Elias Wang, Damian Mrowca, Felix J. Binder, Hsiao-Yu Fish Tung, R. T. Pramod, Cameron Holdaway, Sirui Tao, Kevin A. Smith, Fan-Yun Sun, Li Fei-Fei, Nancy Kanwisher, Joshua B. Tenenbaum, Daniel Yamins, and Judith E. Fan. Physion: Evaluating physical prediction from vision in humans and machines. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2021.

[10] León Begiristain, Olaf Dünkel, and Adam Kortylewski. CRONOS: Benchmarking counterfactual physical consistency in video models. *arXiv preprint arXiv:2605.23699*, 2026.

[11] Florian Bordes, Quentin Garrido, Justine T. Kao, Adina Williams, Michael Rabbat, and Emmanuel Dupoux. IntPhys 2: Benchmarking intuitive physics understanding in complex

synthetic environments. *arXiv preprint arXiv:2506.09849*, 2025.

[12] Emmanuelle Bourigault. World models in words: Auditing physical state-transition commitments in vision-language models. *arXiv preprint arXiv:2605.29585*, 2026.

[13] Shyamal Buch, Cristóbal Eyzaguirre, Adrien Gaidon, Jiajun Wu, Li Fei-Fei, and Juan Carlos Niebles. Revisiting the "video" in video-language understanding. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

[14] Zhenfang Chen, Jiayuan Mao, Jiajun Wu, Kwan-Yee Kenneth Wong, Joshua B. Tenenbaum, and Chuang Gan. Grounding physical concepts of objects and events through dynamic visual reasoning. In *International Conference on Learning Representations (ICLR)*, 2021.

[15] Zhenfang Chen, Kexin Yi, Yunzhu Li, Mingyu Ding, Antonio Torralba, Joshua B. Tenenbaum, and Chuang Gan. ComPhy: Compositional physical reasoning of objects and events from videos. In *International Conference on Learning Representations (ICLR)*, 2022.

[16] Anoop Cherian, Radu Corcodel, Siddarth Jain, and Diego Romeres. LLMPhy: Parameter-identifiable physical reasoning combining large language models and physics engines. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026.

[17] Wei Chow, Jiageng Mao, Boyi Li, Daniel Seita, Vitor Guizilini, and Yue Wang. PhysBench: Benchmarking and enhancing vision-language models for physical world understanding. In *International Conference on Learning Representations (ICLR)*, 2025.

[18] Miles Cranmer, Sam Greydanus, Stephan Hoyer, Peter Battaglia, David Spergel, and Shirley Ho. Lagrangian neural networks. *arXiv preprint arXiv:2003.04630*, 2020.

[19] David Ding, Felix Hill, Adam Santoro, Malcolm Reynolds, and Matt Botvinick. Attention over learned object embeddings enables complex visual reasoning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

[20] Mingyu Ding, Zhenfang Chen, Tao Du, Ping Luo, Joshua B. Tenenbaum, and Chuang Gan. Dynamic visual reasoning by learning differentiable physics models from video and language. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

[21] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman and Hall, 1993.

[22] Aaron Foss, Chloe Evans, Sasha Mitts, Koustuv Sinha, Ammar Rizvi, and Justine T. Kao. CausalVQA: A physically grounded causal reasoning benchmark for video models. *arXiv preprint arXiv:2506.09943*, 2025.

[23] Quentin Garrido, Nicolas Ballas, Mahmoud Assran, Adrien Bardes, Laurent Najman, Michael Rabbat, Emmanuel Dupoux, and Yann LeCun. Intuitive physics understanding emerges from self-supervised pretraining on natural videos. *arXiv preprint arXiv:2502.11831*, 2025.

[24] Tobias Gerstenberg, Noah D. Goodman, David A. Lagnado, and Joshua B. Tenenbaum. A counterfactual simulation model of causal judgments for physical events. *Psychological Review*, 128(5):936–975, 2021.

[25] Sam Greydanus, Misko Dzamba, and Jason Yosinski. Hamiltonian neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[26] Nate Gruver, Marc Finzi, Samuel Stanton, and Andrew Gordon Wilson. Deconstructing the inductive biases of Hamiltonian neural networks. In *International Conference on Learning Representations (ICLR)*, 2022.

[27] Ernst Hairer, Christian Lubich, and Gerhard Wanner. *Geometric Numerical Integration: Structure-Preserving Algorithms for Ordinary Differential Equations*. Springer, 2 edition, 2006.

[28] Joseph Y. Halpern. *Actual Causality*. MIT Press, 2016.

[29] Joseph Y. Halpern and Judea Pearl. Causes and explanations: A structural-model approach. Part I: Causes. *The British Journal for the Philosophy of Science*, 56(4):843–887, 2005.

[30] Kaiming He, Georgia Gkioxari, Piotr Dollár, and Ross Girshick. Mask R-CNN. In *IEEE International Conference on Computer Vision (ICCV)*, 2017.

[31] Adam Ishay, Zhun Yang, Joohyung Lee, Ilgu Kang, and Dongjae Lim. Think before you simulate: Symbolic reasoning to orchestrate neural computation for counterfactual question answering. In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2024.

[32] Steeven Janny, Fabien Baradel, Natalia Neverova, Madiha Nadri, Greg Mori, and Christian Wolf. Filtered-CoPhy: Unsupervised learning of counterfactual physics in pixel space. In *International Conference on Learning Representations (ICLR)*, 2022.

[33] Bingyi Kang, Yang Yue, Rui Lu, Zhijie Lin, Yang Zhao, Kaixin Wang, Gao Huang, and Jiashi Feng. How far is video generation from world model: A physical law perspective. In *International Conference on Machine Learning (ICML)*, 2025.

[34] Chia-Hsiang Kao, Cong Phuoc Huynh, Chien-Yi Wang, Noranart Vesdapunt, Stefan Stojanov, Bharath Hariharan, Oleksandr Obiednikov, and Ning Zhou. Δynamics: Language-based representation for inferring rigid-body dynamics from videos. *arXiv preprint arXiv:2605.20576*, 2026.

[35] Benno Krojer, Mojtaba Komeili, Candace Ross, Quentin Garrido, Koustuv Sinha, Nicolas Ballas, and Mahmoud Assran. A shortcut-aware video-QA benchmark for physical

understanding via minimal video pairs. *Transactions on Machine Learning Research*, 2025.

[36] David Lewis. Causation. *The Journal of Philosophy*, 70(17): 556–567, 1973.

[37] Yunzhu Li, Jiajun Wu, Jun-Yan Zhu, Joshua B. Tenenbaum, Antonio Torralba, and Russ Tedrake. Propagation networks for model-based control under partial observation. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2019.

[38] Ruibo Liu, Jason Wei, Shixiang Shane Gu, Te-Yen Wu, Soroush Vosoughi, Claire Cui, Denny Zhou, and Andrew M. Dai. Mind's eye: Grounded language model reasoning through simulation. In *International Conference on Learning Representations (ICLR)*, 2023.

[39] Jiayuan Mao, Xuelin Yang, Xikun Zhang, Noah D. Goodman, and Jiajun Wu. CLEVRER-Humans: Describing physical and causal events the human way. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2022.

[40] Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, 1947.

[41] Andrew Melnik, Robin Schiewer, Moritz Lange, Andrei Muresanu, Mozhgan Saeidi, Animesh Garg, and Helge Ritter. Benchmarks for physical reasoning AI. *Transactions on Machine Learning Research*, 2023.

[42] Brian Mirtich and John Canny. Impulse-based simulation of rigid bodies. In *Proceedings of the 1995 Symposium on Interactive 3D Graphics*, pages 181–188, 1995.

[43] Saman Motamed, Laura Culp, Kevin Swersky, Priyank Jaini, and Robert Geirhos. Do generative video models understand physical principles? In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2026.

[44] Nick Pawlowski, Daniel C. Castro, and Ben Glocker. Deep structural causal models for tractable counterfactual inference. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[45] Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2 edition, 2009.

[46] Tim Rädsch, Yuki M. Asano, Hilde Kuehne, Stefan Bauer, Priyank Jaini, Robert Geirhos, and Carsten T. Lüth. Physics-IQ verified. *arXiv preprint arXiv:2606.18943*, 2026.

[47] Fabio Rovai. Deterministic event-graph substrates as world models for counterfactual reasoning. *arXiv preprint arXiv:2605.15967*, 2026.

[48] Alvaro Sanchez-Gonzalez, Victor Bapst, Kyle Cranmer, and Peter Battaglia. Hamiltonian graph networks with ODE integrators. *arXiv preprint arXiv:1909.12790*, 2019.

[49] Alvaro Sanchez-Gonzalez, Jonathan Godwin, Tobias Pfaff, Rex Ying, Jure Leskovec, and Peter W. Battaglia. Learning to simulate complex physics with graph networks. In *International Conference on Machine Learning (ICML)*, 2020.

[50] Bernhard Schölkopf, Francesco Locatello, Stefan Bauer, Nan Rosemary Ke, Nal Kalchbrenner, Anirudh Goyal, and Yoshua Bengio. Toward causal representation learning. *Proceedings of the IEEE*, 109(5):612–634, 2021.

[51] Vinay Sharma and Olga Fink. A physics-informed graph neural network conserving linear and angular momentum for dynamical systems. *Nature Communications*, 17, 2026.

[52] Andrew Sosanya and Sam Greydanus. Dissipative hamiltonian neural networks: Learning dissipative and conservative dynamics separately. *arXiv preprint arXiv:2201.10085*, 2022.

[53] Hsiao-Yu Fish Tung, Mingyu Ding, Zhenfang Chen, Daniel M. Bear, Chuang Gan, Joshua B. Tenenbaum, Daniel Yamins, Judith E. Fan, and Kevin A. Smith. Physion++: Evaluating physical scene understanding that requires online inference of different physical properties. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2023.

[54] Zhekai Wang, Haoxiang Huang, Xiang Liu, Zhikang Chen, Yueqing Sun, Qi Gu, Shiji Zhou, Miao Liu, and Sen Cui. MOSH-WM: Mask-grounded soft-Hamiltonian dynamics for object-centric world models. *arXiv preprint arXiv:2608.22750*, 2026.

[55] Jiajun Wu, Ilker Yildirim, Joseph J. Lim, William T. Freeman, and Joshua B. Tenenbaum. Galileo: Perceiving physical object properties by integrating a physics engine with deep learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2015.

[56] Jiajun Wu, Erika Lu, Pushmeet Kohli, William T. Freeman, and Joshua B. Tenenbaum. Learning to see physics via visual de-animation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[57] Ziyi Wu, Nikita Dvornik, Klaus Greff, Thomas Kipf, and Animesh Garg. SlotFormer: Unsupervised visual dynamics simulation with object-centric models. In *International Conference on Learning Representations (ICLR)*, 2023.

[58] Chen Yang, Shenxiang Zeng, Haoyang Zhao, Zhouyuan Xu, Youquan He, Haoyu Li, Mingyi Deng, Jiansheng Fan, and Chen Wang. PhysMind: From video to executable worlds for training-free physical reasoning. *arXiv preprint arXiv:2608.04575*, 2026.

[59] Kexin Yi, Chuang Gan, Yunzhu Li, Pushmeet Kohli, Jiajun Wu, Antonio Torralba, and Joshua B. Tenenbaum. CLEVRER: Collision events for video representation and reasoning. In *International Conference on Learning Representations (ICLR)*, 2020.

A Protocol

Pre-registration. Before the fresh test set was scored, we wrote down the inputs, the sets, the selection rule, the systems to be scored and the statistics to be reported. Settings were chosen on VAL-A only; TEST-A and TEST-B were then scored once per listed system, and nothing was changed afterwards. The listed systems were the learned engine, the analytic rigid-body, straight lines, straight lines with friction, no simulation, seven one-change ablations of the learned engine, nine other checkpoints of the learned model, six retrained seeds, a vision-language baseline, the other three question types, and the learned engine with the dataset's own question programs as a check of the parser. Anything else reported in the paper is labelled as a separate analysis (the evidence audit, the explanatory but-for test and the key-consistency check).

History of the sets. TEST-A served as the test set of earlier versions of the engine, so some design choices may have been influenced by it. For this reason we added TEST-B, 898 CLEVRER training videos that no model had been trained on and that had never been scored, and named it the primary test set. Every significant comparison of the main paper keeps its direction on TEST-B alone (Section J); the entry correction, not significant on the pooled sets, is significant on TEST-B alone ($z = -2.2$).

Selection on VAL-A. For each simulator we swept the hand-off look-ahead $L \in \{1, 3, 6, 12\}$, five collision detectors and the extension $E \in \{0, 10, 20, 30, 45, 60\}$, and kept the setting with the most correct options (ties broken by correct questions). Table S1 lists the chosen settings.

Table S1: Settings chosen on VAL-A (941 questions, 3,391 options). Union: calibrated distance rule OR path change.

Simulator	Look-ahead L	Detector, extension E	Options	Questions
analytic rigid-body	3	union, +60	91.5 (3,104)	74.9 (705)
learned world model	3	union, +30	90.3 (3,061)	71.2 (670)
straight + friction	1	union, +45	89.9 (3,047)	71.0 (668)
straight lines	1	union, +45	89.3 (3,028)	69.2 (651)
no simulation	none	union	81.7 (2,769)	50.3 (473)

Statistics. Intervals resample videos, because the options of one question and the questions of one video are not independent: 2,000 resamples with a fixed seed. Paired comparisons use McNemar's test on the options both systems answer, reported as $z = (b - c)/\sqrt{b + c}$, where b and c count the options only one of the two systems gets right; the same is done for questions. For the pooled sets the set name is part of the video identifier, so that the bootstrap and the grouping of questions stay correct.

B Perception and question parsing

Tracks from detections. We use the Mask R-CNN detections released with NS-DR for all 20,000 CLEVRER videos, each with predicted colour, material and shape. Detections of the same colour and shape closer than 8 pixels are merged; two objects may share colour and shape only if two detections appear at least 20 pixels apart in 3 or more frames; tracks shorter than 5 frames are dropped; gaps of up to 5 frames are interpolated; the material is a score-weighted majority vote. Objects are identified by colour, material and shape together, because 21% of CLEVRER videos contain two objects that share colour and shape.

Camera. The camera published for CLEVR scenes does not match CLEVRER. We recovered it by rendering the convex hulls of the three shapes and fitting camera position and focal length to the detected masks, which raised the median silhouette IoU from 0.57 to 0.93. Mask centres are lifted to the table plane at the object's centre height.

Tracking quality. Table S2 compares the object inventory of each video with the annotation, which is used only for this check.

Table S2: Detection-only tracks against the annotation (object inventory per video).

Set	Videos	Recall	Precision	Exact inventory	Frames in view co
VAL-A	500	99.4	99.7	95.8	98.6
TEST-A	500	99.8	99.7	97.6	98.5
TEST-B	898	99.6	99.4	95.2	98.6

Question parser. Questions are parsed by templates learned from CLEVRER's training questions: each template has typed slots that are filled by voting over the training programs. On the validation questions the parser reproduces the dataset's programs exactly for 100% of counterfactual and predictive questions, 99.995% of explanatory questions and 99.80% of descriptive questions. With the dataset's programs instead of parsed ones, the scores change by at most 0.2 points.

C Model, settings and compute

Training. The base model was trained for 30 epochs on the tracks of 11,182 training videos with a rollout loss

Table S3: Learned world model (43,460 trained parameters).

Part	Shape	Parameters
attribute encoder	$8 \to 128 \to 64$	9,408
mass head	$64 \to 128 \to 1$	8,449
pair energy V_θ	$68 \to 128 \to 128 \to 1$	25,473
translational friction head	$64 \to 1$	65
rotational friction head	$64 \to 1$	65
total		43,460

whose horizon grew from 3 to 20 frames (AdamW, learning rate 10^{-3}, batch 16, gradient clipping at 1.0; 5.2 hours). The final model was fine-tuned for 4 epochs at horizon 20 with voxel contacts inside the rollout (learning rate 3×10^{-4}; 2.9 hours), keeping the epoch with the lowest validation rollout error. Object radii are measured from the masks and kept fixed. Six seeds repeat the fine-tune. Seeds 1 and 2 were stopped after two of four epochs by a power loss, and in both the best epoch was the first, as in the original run; seeds 3 to 6 ran all four epochs on a free hosted CPU session and their best epochs were the fourth, second, third and third. Their validation rollout errors at 20 frames are 0.0214, 0.0253, 0.0231 and 0.0239 against the released model's 0.0230, but the trainer draws its validation windows from the seed, so these are not measured on identical windows; relative to its own starting point seed 3 improves 22% where the released model improved 13%. Across the five checkpoints that have a validation number the correlation between rollout error and options correct is -0.76, with the benchmark spanning 0.5 points while the rollout error spans 18%. The seven checkpoints of this recipe score 90.3 ± 0.2 per option and 71.8 ± 0.6 per question.

Vision-language baseline. We also asked a general vision-language model the same options, to see what the task looks like to a system that watches the video and answers from what it sees. Qwen2.5-VL-7B-Instruct receives 16 frames spread evenly over the clip and the list of events, and is asked whether each event happens once the named object is removed; decoding is greedy, and the polarity of questions that ask which event will *not* happen is applied when scoring, so the number does not depend on how the model handles negation. The same run was repeated with the frames withheld.

With the frames the model is indistinguishable from never predicting a collision (54.4 against 54.1), and it is 35 points below the engine on the same options (paired: the engine is right where the model is wrong 1,362 times, the reverse 177 times, $z = -30.2$). It answers "no" to every event of 86% of questions when it sees the frames and "yes" to every event of 70% of them when it does not, so what changes its answer is whether images are present rather than which object was removed. Two of 1,859 replies could not be read and were counted as "no". Concurrent work measures the same effect on a different subset and with much larger models: a blind vision-language model answers 50.1 of counterfactual options, GPT-4o 56.5 and GPT-5.5 76.7, against a 50.0 chance level (PhysMind, Table 1, cited in the main paper). Our 54.4 for a 7B open model sits where that range predicts, and the blind number is close to the 48.8 we measure. This is a baseline and not a fair comparison with the published systems: it is one prompt of one open model with no fine-tuning on this dataset, and a system built around such a model could do better. It does say that the frames alone, read by a strong general model, do not answer these questions.

Table S4: A vision-language model on the 3,332 options of TEST-A, options / questions correct (%), with the two policies that ignore the question and the engine for reference. The last column is how often the model answers that an event happens.

System	Option	Question	Says yes
answer "no" to every event	54.1	0.0	0
Qwen2.5-VL-7B, 16 frames	54.4	2.9	9
Qwen2.5-VL-7B, no frames	48.8	4.2	78
answer "yes" to every event	45.9	2.6	100
CausalVis, learned model	89.9	70.8	–

Analytic rigid-body simulator. Fitted on training tracks: cubes and cylinders slide with a constant deceleration of 0.00008 per frame; spheres roll and lose 0.3% (metal) or 0.2% (rubber) of their speed per frame; collisions apply an impulse along the line of centres with restitution 0.2, a metal to rubber mass ratio of 1.00 and contact friction 0.05.

Engine. Positions are in scene units divided by 6. The calibrated distance rule counts a collision when the centre distance is below $1.2(r_i + r_j) + 0.03$; the path-change rule when the surfaces are within 0.03 and one object's velocity changes by at least 0.002 within 6 frames; contacts closer than 6 frames are merged. An object that leaves the table ($|x| > 0.95$) stops interacting.

Other question types. Descriptive: calibrated rule with the collision onset moved to the closest approach within 6 frames; an object is moving if its speed exceeds 0.001. Explanatory: the same detector and CLEVRER's chain rule. Predictive: the plain distance rule ($r_i + r_j + 0.02$) and 60 frames of simulation past the end. All chosen on VAL-A.

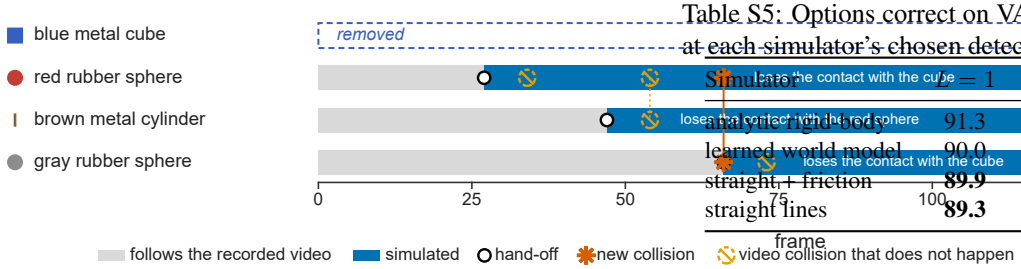


Figure S1: The engine on the example of Figure 1 in the main paper (video 10880, the blue cube removed). Each object follows the recorded video until the removal can reach it, and is handed to the simulator 3 frames before it would lose a recorded contact. The red sphere then misses the cylinder and strikes the gray sphere at frame 66.

Compute. One laptop CPU (Intel i5-8350U, 4 cores, 8 threads, 7.9 GB of memory), no GPU, for every stage. Answering one counterfactual question (building its worlds and reading every option) takes 2.24 seconds with the learned model, 1.36 with the rigid-body simulator, 1.40 with straight lines and 1.04 without physics, measured on 40 TEST-A questions with one process, one thread and nothing else running (start-up excluded). Training took 8.1 hours in total.

D Hand-off look-ahead and extension

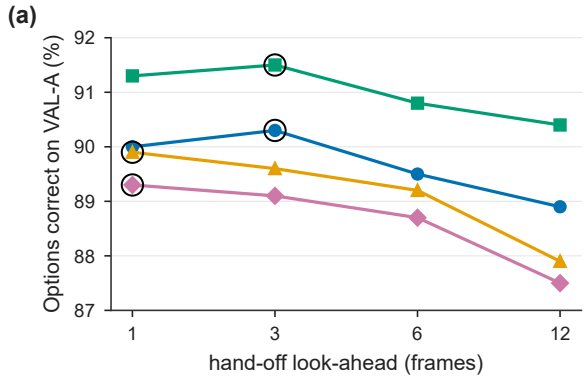


Figure S2: Settings chosen on VAL-A. (a) Handing an object to the simulator later helps every simulator. (b) Simulating past the video end adds 4 to 5 points, because CLEVRER's answers include collisions after the last frame.

Table S5: Options correct on VAL-A (%) by look-ahead L, at each simulator's chosen detector and extension.

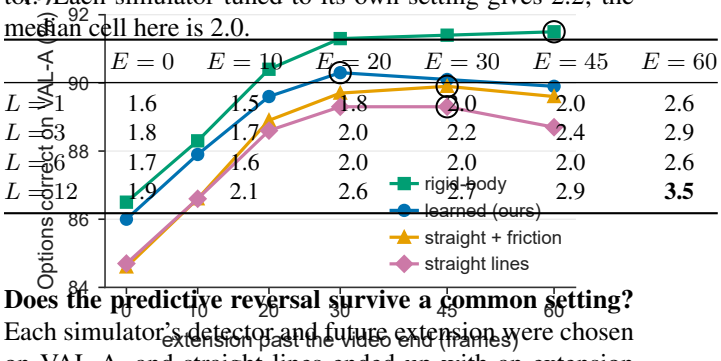
Simulator	$L = 1$	$L = 3$	$L = 6$	$L = 12$
rigid-body	91.3	**91.5**	90.8	90.4
learned world model	90.0	**90.3**	89.5	88.9
straight + friction	**89.9**	89.6	89.2	87.9
straight lines	**89.3**	89.1	88.7	87.5

Is the simulator spread an artefact of tuning each simulator separately? Each simulator is given its own look-ahead and extension, chosen on VAL-A, and the extension alone is worth 4.6 points, so a weak simulator allowed to simulate less could look better than it is. Table S6 answers this directly: it holds all four simulators at the same setting and reports how far apart they then fall. Tuning does not compress the range. Each tuned to its own setting they differ by 2.2 points, and the median common setting gives 2.0.

What the table does show is that the spread depends on the regime. It is smallest when objects are handed over late and the world is not extended, and largest at $L = 12$ with 60 extra frames, where the engine simulates most. The benchmark becomes more sensitive to the simulator the more of the world a system has to simulate, and the best-scoring configuration is the one that simulates least. The insensitivity we report is a property of that regime, not of counterfactual questions in general.

Table S6: Spread between the four simulators, in points of options correct, with all four held at the same look-ahead L and extension E. VAL-A, 3,391 options, union detector. Each simulator tuned to its own setting gives 2.2; the median cell here is 2.0.

	$E = 0$	$E = 10$	$E = 20$	$E = 30$	$E = 45$	$E = 60$
$L = 1$	1.6	1.5	1.8	2.0	2.0	2.6
$L = 3$	1.8	1.7	2.0	2.2	2.4	2.9
$L = 6$	1.7	1.6	2.0	2.0	2.0	2.6
$L = 12$	1.9	2.1	2.6	2.7	2.9	**3.5**



Does the predictive reversal survive a common setting? Each simulator's detector and future extension were chosen on VAL-A, and straight lines ended up with an extension of 45 frames where the other three got 60, so the published predictive ranking is not a single-variable comparison. Table S7 re-scores the stored predictive runs at every common setting.

The ranking is not a property of the simulators. Under the distance rule the analytic rigid-body simulator comes last at every extension, which is the reversal the paper reports; under the calibrated rule or the union they are never

last and are first at the longer extensions. Counting over the fifteen settings, straight lines rank first in 11 and the rigid-body simulator in 4 on test, and 8 and 5 on VAL-A. The `kick2` rows are left out of those counts: that detector fires on a change of velocity, so for straight-line motion it reports nothing at all and returns exactly the 50.0% that answering "no" to every option scores, which is an artefact of the detector rather than a measurement of the simulator.

Table S7: Predictive options correct (%) on TEST-A and TEST-B, all four simulators held at the same detector and extension. 1,996 options. The rigid-body simulator has the lowest rollout error of the four.

Detector	E	straight	str+fric	learned	laws
rule	30	**88.3**	85.8	85.8	83.0
rule	45	**91.4**	90.5	90.5	87.5
rule	60	**91.4**	90.9	90.5	88.1
cal	30	**86.2**	85.5	85.1	85.6
cal	45	87.0	87.6	87.1	**88.0**
cal	60	86.9	87.3	86.4	**87.9**
union	30	**86.2**	85.5	85.8	86.0
union	45	87.0	87.6	87.7	**88.4**
union	60	86.9	87.3	87.1	**88.3**

An oracle event detector. On 52.2% of test options neither asked object is ever handed to the simulator, so the counterfactual world is identical to the recording for that pair and a perfect detector is exactly CLEVRER's collision annotation. Table S8 substitutes it there and leaves everything else alone. It is a diagnostic: no setting is chosen on it.

The substitution touches only the options it applies to, which is the check that it was implemented correctly: on those, accuracy rises from 93.0% to 95.7%, while the simulated pairs stay at 87.6% exactly. Even with perfect detection inside the video those pairs stop at 95.7%, and 44% of what remains is the answer key asking for a collision the video never shows.

Perfect detection on half the benchmark is worth 1.4 to 1.5 points, close to what the whole 2.2× sweep of physical accuracy is worth, and the ceiling of 92.9% is still 3.2 points below CRCG. The third row repeats the substitution with the oracle blind to everything after the last frame: it drops to 88.2%, below the engine itself. A perfect detector that only sees the video loses to an imperfect detector allowed to simulate past it, which is the post-video finding of the main paper established without any model.

Table S8: An oracle detector on every pair the engine never simulates. Options correct [questions correct], with McNemar z against the unmodified engine on the same options.

	learned		analytic rigid-body	
Detection on the recorded pairs	Opt.	Ques.	Opt.	Ques.
the engine's own rule	90.4	72.3	91.4	75.0
oracle, engine still extends past the end	**91.8**	**76.6**	**92.9**	**79.6**
McNemar z	$+9.8$		$+10.2$	
oracle, nothing after the last frame	88.2	64.5	89.0	66.5
McNemar z	-8.1		-8.4	

E Question subsets and errors

Following ALOE (Appendix C of Ding et al., 2021), a question belongs to E if the removed object never collides in the video, to M if it collides but the recorded events already answer every option, and to H otherwise.

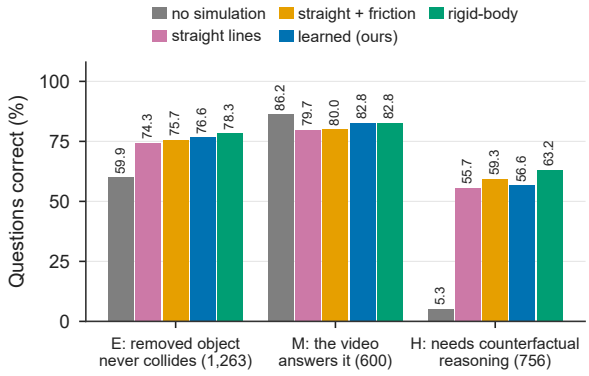


Figure S3: Where physics matters. Without physics the H questions fall to 5.3%; any simulator recovers most of them, and the simulators differ by a few points.

Table S9: Subsets on TEST-A and TEST-B, options / questions correct (%).

System	E (1,263 q)	M (600 q)	H (756 q)
no simulation	87.5 / 59.9	95.6 / 86.2	60.0 / 5.3
straight lines	91.5 / 74.3	93.1 / 79.7	83.9 / 55.7
straight + friction	91.9 / 75.7	93.3 / 80.0	84.9 / 59.3
learned world model	92.6 / 76.6	94.4 / 82.8	83.6 / 56.6
analytic rigid-body	93.2 / 78.3	94.4 / 82.8	86.0 / 63.2

In E, the answer key says that a pair collides on 474 of 4,688 options (10.1%) where the video contains no collision of that pair; these options occur in 437 of 1,263 questions (34.6%). Since nothing in the video changes when

the removed object never collides, these collisions can only happen after the last frame.

Errors of the learned engine (896 wrong options): a collision invented during the video 360 (40%); a collision missed after the video ends 274 (31%); a collision invented after the video ends 139 (16%); a visible collision missed 89 (10%); an object not found by the detector 34 (4%).

What a wrong answer looks like. The engine stores the deciding event for every option, so the evidence behind an error can be read directly. Table S10 lists one case of each kind, taken from the test records (`paper_exp/failure_examples.py` writes all of them). Grouping the 896 wrong options of the learned system this way gives 360 collisions invented inside the video, 308 collisions the key places after the last frame that the extension does not reach, 139 invented in the extension, and 89 annotated collisions missed outright.

Two things are visible in the table. The invented cases are mostly pairs that never leave the recording, so no simulator was involved and the detector alone is responsible; this is the same conclusion the oracle detector reaches by a different route. The missed cases divide sharply: only 89 of 896 are collisions actually visible in the video, and three and a half times as many are events the answer key places after the recording ends.

Table S10: One wrong answer of each kind, with the evidence the engine stored. "Fired" is the frame at which our detector reported contact, "annotated" is CLEVRER's collision annotation for the same pair, and the video is 128 frames, so a frame above 127 lies in the extension.

Kind	n	Video	Pair state	Fired	Annotated
invented in video	360	150	on the recording	30	none
invented after end	139	820	on the recording	143	none
missed after end	308	290	on the recording	never	none
missed, visible	89	7160	simulated	never	72

F Evidence audit

An answer's deciding event is the first collision of the asked pair in the counterfactual world, or its absence. It is *recorded* when both objects still follow the recording at that frame and *simulated* otherwise. A recorded collision is confirmed if the annotation has a collision of the same pair within a tolerance; a recorded absence is confirmed if the annotation has no collision of the pair in the whole video (the part after the video end is simulated and not checked).

The deciding recorded collision lies a median of 4 frames before the annotated one (10th percentile -7, 90th percentile -2; 2,607 answers), because the calibrated rule fires when surfaces enter its margin. Over all recorded videos, the detector used for the other question types (calibrated rule with onset refinement) finds 3,229 of the 3,396 annotated collisions (recall 0.95) and reports 869 others (precision 0.79), with a median timing offset of 0 frames.

Table S11: Evidence behind the answers of the learned engine (TEST-A and TEST-B).

Correct answers (8,454)	15-frame tolerance	6-fr
recorded collision, confirmed	2,567	
recorded absence, confirmed	2,314	
recorded, annotated collision of the pair further away	40	
recorded, no annotated collision of the pair	52	
simulated	3,426	
asked object removed or not found	55	
confirmed share of checkable	98.2%	
Wrong answers (896)		
detector reports a collision the annotation lacks	188	
detector misses an annotated collision	37	
key needs a collision after the video ends	72	
other recorded cases	8	
simulated	557	
object not found by the detector	34	

G Explanatory questions and CLEVRER-Humans

But-for test on CLEVRER. For each option we locate the target collision and the candidate cause on the recorded events. A candidate "presence of Z" or "Z's entrance" removes Z; a candidate collision removes whichever of its objects is not in the target collision. A candidate that shares both objects with the target cannot be removed without removing the target and keeps the chain-rule answer, as do candidates that could not be resolved. The extension past the video end for the target check was chosen on VAL-A (+0 frames: 76.8%; +10: 76.4%; +30: 75.3%).

Table S12: Explanatory options on TEST-A and TEST-B (8,566 options), by kind of candidate cause.

Candidate	Options	But-for test	Chain rule
another collision	2,389	78.3	98.1
an entrance	1,321	88.2	97.2
the presence of an object	3,292	85.7	95.6
not resolvable (same answer)	1,564	51.0	51.0
all	8,566	77.7	88.4

Key consistency. Table S13 matches explanatory options about the presence or entrance of Z with counterfactual options about the same pair without Z in the same video. No simulator is involved.

Table S13: CLEVRER's explanatory key against its own counterfactual key.

Set	Matched pairs	"Responsible", yet the pair still collides without Z	"Not responsible", yet the pair no longer collides without Z
VAL-A	446	37 / 87 (42.5%)	1 / 359 (0.3%)
TEST-A	423	27 / 85 (31.8%)	0 / 338 (0.0%)
TEST-B	817	49 / 148 (33.1%)	0 / 669 (0.0%)
all	1,686	113 / 320 (35.3%)	1 / 1,366 (0.1%)

Is it the same collision? CLEVRER's counterfactual answer carries no time, so "X and Y still collide without Z" could refer to a different collision. For the 113 cases in which the key calls Z responsible and the pair still collides without Z, the pair collides exactly once in the recorded video in all 113. In the engine's world without Z, the first collision of the pair lies within 10 frames of the recorded one in 56 cases, more than 10 frames earlier in 23, more than 10 frames later in 14, and does not happen in 20.

CLEVRER-Humans. CLEVRER-Humans describes events in free-form sentences. We linked each sentence to objects and a frame with a language model (gpt-oss-120b through the Groq service, temperature 0, outputs cached and released), then asked whether the cause event is responsible for the effect event. The but-for test removes the cause's object and checks whether the effect still happens in the simulated world; the chain rule applies CLEVRER's rule to our recorded events. The rule was chosen on 102 grounded training videos (240 options: but-for 64.6%, chain rule 58.8%, always yes 52.1%). On the validation split (216 questions, 432 options) the but-for test answers 64.4% of options and 43.5% of questions, the chain rule 60.9% and 39.4%, always yes 50.7% and 30.1%, the best published model 54.0% and 34.2%, and people 84.5% and 71.4%. The two rules disagree on 93 of the 432 options, 54 of them in but-for's favour, so McNemar gives $z = 1.56$: on this set the margin is not significant, and we report it as suggestive only. The model-free comparison between CLEVRER's own two answer keys above does not depend on it. Of the 183 validation videos, 150 lie in our world model's training split; only their trajectories were used, never any label.

H ComPhy

ComPhy hides mass and charge. Each target video comes with four reference videos of the same objects. We take object tracks from the annotation (visible frames of the first five seconds only), infer each object's mass (light or heavy) and charge (positive, negative or none) by searching for the assignment whose simulations best fit the target and reference videos, and simulate interventions and the future with laws fitted on training annotations (friction, charge constant of 0.138, contact and restitution). Settings were chosen on 300 training scenes: reference videos on, counterfactual extension +0 frames, predictive extension +50 frames. The hidden mass is recovered exactly in 98% of mass scenes; whether an object is charged in 99.6% of charge scenes, but the exact signs of all charges in only 11%. The validation set has 2 options per question and the test set mostly 4, so per-question accuracy is not comparable; ours is 66.7% (counterfactual) and 74.1% (predictive). 371 of 15,968 options name an object we could not resolve and are answered no.

Evidence audit on ComPhy. ComPhy's annotation includes the frames after the end of each video (frames 125 to 174), so the event that decides a predictive answer can be checked. On 400 validation scenes (1,597 predictive options whose two objects we resolve), 1,411 answers are correct. Of these, 589 are decided by a simulated collision that matches a real future collision of the same pair within 5 frames (553 within 3 frames, 603 within 10), 789 by an absence of collision that matches the real future, and 33 by a simulated collision with no real counterpart within 5 frames: 97.7% of the correct answers rest on a real event. Of the 186 wrong answers, 152 miss a real future collision, 8 invent one, and 26 say no collision where the annotated future window also shows none, which suggests collisions after frame 174 in the answer key. Counted per asked pair, the first simulated future collision is real in 93.5% of cases (589 of 630) and finds 76.9% of the real ones (589 of 766).

Table S14: ComPhy, accuracy per option. Ours: validation set (2,000 scenes) with oracle tracks; published: test set.

Method	Counterfactual	Predictiv
ALOE (reference videos)	67.9	67.9
NS-DR+ (given properties)	61.1	73.3
CPL	68.3	75.3
CPL-Gt (ground-truth properties and dynamics)	74.0	87.6
CausalVis engine (oracle tracks, inferred properties)	79.5	86.9
people	80.0	88.0

I CoPhy

On BallsCF, the engine receives the object states of the observed run AB and the new initial state C, infers each ball's mass and restitution from AB by simulation, and predicts the counterfactual run D with planar laws fitted on the 4-ball training split. The metric is the squared 2D position error summed over x and y and averaged over time steps and balls.

Table S15: CoPhy BallsCF test splits (1,000 scenes each), counterfactual position error (lower is better).

Balls	2	3	4	5	6
CausalVis engine, inferred confounders	0.89	1.56	2.03	2.39	2.60
same engine, true confounders	0.42	0.72	1.09	1.28	1.55
copy the observed trajectory (B)	7.69	6.81	6.18	5.93	5.74
copy the new initial state (C)	12.58	12.22	11.88	11.65	11.26

The released CoPhy files differ from the description in the paper (masses 1, 2 or 5 and restitution 0.1, 0.5 or 1), and no variant of the metric reproduces the published copy baselines, so published CoPhyNet numbers are not comparable with ours. Event audit (300 test scenes per split, same pair within one step): 2 balls, precision 0.85 and recall 0.65 (64 / 11 / 35 true positives, false positives, misses); 4 balls, 0.61 and 0.49 (279 / 180 / 294); 6 balls, 0.49 and 0.41 (631 / 651 / 910). CollisionCF and BlocktowerCF were not run, because their objects fall and topple, which the planar engine does not model.

Table S16: Counterfactual questions, options / questions correct (%), with paired z against the learned engine on options. One change at a time below the line.

System	TEST-A	TEST-B
learned world model (CausalVis)	90.2 / 71.4	90.5 / 72.8
analytic rigid-body	91.7 / 75.3 ($+4.2$)	91.3 / 74.8 ($+3.1$)
straight + friction	90.1 / 71.4 (-0.2)	90.2 / 72.2 (-0.9)
straight lines	89.7 / 70.2 (-1.0)	89.6 / 70.2 (-2.7)
no simulation	81.3 / 49.4 (-13.1)	81.5 / 50.6 (-18.4)
no extension past the video end	86.2 / 58.9 (-8.6)	85.7 / 57.5 (-13.6)
12-frame hand-off	88.1 / 66.1 (-5.2)	89.8 / 70.8 (-2.8)
distance rule only	89.9 / 70.8 (-2.1)	90.2 / 71.9 (-4.2)
no voxel contact impulse	89.8 / 70.8 (-1.2)	89.6 / 70.1 (-3.7)
no learned pair energy	90.5 / 72.2 ($+1.3$)	91.0 / 74.1 ($+3.0$)
no learned friction	90.9 / 73.2 ($+2.1$)	90.3 / 71.7 (-0.9)
no entry correction	90.3 / 71.3 ($+0.8$)	90.4 / 72.4 (-2.2)
seed 1 (2 of 4 epochs)	90.3 / 71.6 ($+0.5$)	90.6 / 72.8 ($+0.6$)
seed 2 (2 of 4 epochs)	90.1 / 71.0 (-0.5)	90.1 / 71.5 (-2.4)
seed 3	90.3 / 71.3	90.6 / 73.0
seed 4	90.1 / 71.1	90.2 / 71.9
seed 5	90.3 / 71.0	90.4 / 72.6
seed 6	90.2 / 71.0	89.9 / 70.8

J Full results per test set

Table S17: Other checkpoints of the learned model in the same engine, options / questions correct (%), and their rollout error at 40 frames on VAL-A.

Checkpoint	Rollout error	TEST-B	TEST-A
3D data, drag	0.067	89.8 / 70.6	89.9 / 70
radius frozen	0.071	89.8 / 70.8	89.9 / 70
contact impulse in training	0.064	90.9 / 73.8	90.9 / 73
first spatial model (about 3,200 videos)	0.070	90.0 / 71.2	90.2 / 71
base model before the voxel fine-tune	0.076	89.2 / 69.2	89.2 / 69
fine-tune without voxels	0.067	90.6 / 73.2	90.6 / 72
long rollouts, fixed length	0.062	90.5 / 72.3	90.3 / 71
long rollouts, curriculum	0.064	90.2 / 71.4	90.2 / 71
long rollouts, control	0.062	90.6 / 73.2	90.4 / 72